Zecheng(Zephyr) Yin

Shenzhen, China | yonggiee@qq.com | +86 17728153439 | yonggie.github.io

linkedin.com/in/zecheng-yin-95ba8416a | github.com/yonggie

Hard-working bee

Hi. My name is Zecheng (Zephyr) Yin. I'm a fanatic about cutting-edge research on Embodied AI (Vision Language Navigation & Vision Language Action), Natural Language Processing (NLP), VLM/LLM, Computer Vision, and Graph Data Mining. I am currently a researcher and engineer at Shenzhen Future Network of Intelligence Institute (FNii) led by Shuguang Cui, a Fellow of the Canadian Academy of Engineering, and working closely with Prof. Zhen Li at The Chinese University of Hong Kong (Shenzhen).

During my M.S., I was advised by Prof. Yanchun Zhang and Hong Yang in the area of graph neural networks, NLP data mining, and contrastive learning at Guangzhou University. During this period, I had an internship at Kuaishou for NLP search mining and an internship at IDEA for financial graph malware detection. Prior to my M.S., I was advised by Prof. Jin Li in the area of federated learning and attack at Guangzhou University during my B.S.

Publications

EffiNav: Efficient Object Goal Navigation via VLM-Depth-Guided Exploration in Unknown Environments — Mar 2026
Zecheng Yin, Shuguang Cui, Zhen Li

HyPerNav: Hybrid Perception for Object-Oriented Navigation in Unknown Environments, IROS'26 (Under Review) — Sept 2025
Zecheng Yin, Shuguang Cui, Zhen Li

Navigation with VLM Framework: Go to Any Language, IROS'26 (Under Review) — Sept 2024
Zecheng Yin, Chonghao Cheng, Yao Guo, Zhen Li

TCMCoRep: Traditional Chinese Medicine Data Mining with Contrastive Graph Representation Learning, KSEM'23 (CCF-C, acceptance rate: 23.1%) — May 2023
Zecheng Yin, Jinyuan Luo, Yuejun Tan, Yanchun Zhang

HGCL: Heterogeneous Graph Contrastive Learning for Traditional Chinese Medicine Prescription Generation, HIS'22 (acceptance rate: 27.78%) — Aug 2022
Zecheng Yin, Yingpei Wu, Yanchun Zhang

A Hybrid-Scales Graph Contrastive Learning Framework for Discovering Regularities in Traditional Chinese Medicine Formula, BIBM'21 (CCF-B, acceptance rate: 19.6%) — Feb 2021
Yingpei Wu, ***Zecheng Yin***, Kaiyuan Zhou, Ruofei Wang, Yun Yang, Zepeng Yin, Yanchun Zhang

Education

Guangzhou University, M.S. in Computer Science — Sept 2020 – May 2023
- GPA: 3.4/4.0

Guangzhou University, B.S. in Computer Science — Sept 2016 – May 2020
- GPA: 3.36/4.0

University of Washington, AI & Robotics Program — Aug 2018 – Sept 2018
- AI & Robotics Program, Certificate of Excellence, mentored by Melody Su

Proficiency

3D Perception / SLAM
- RGB-D, camera intrinsics, extrinsics, pose acquirement, active transformation, passive transformation.
- Spatial unification. Coordinate calibration, path planning & following.
- 3D reconstruction, point cloud, voxelization, camera.

Artificial Intelligence

- VLM fine-tuning, development (vLLM, SGLang).
- PyTorch structure modification.
- Implementation of self-designed QwenVL/InternVL server via Flask.
- LoRA PPO reinforcement learning fine-tuning on minicpm-v-2.5.

Simulation and Real-World Robots Development

- Simulation environment such as Habitat, and Omnigibson (BEHAVIOR-1K).
- Simulation robot control, e.g. arm manipulation (IK-controller) and locomotion.
- Simulation environment scene design.
- Real-world robots communication, implementation, and deployment.
- Real-world robot ROS/C++/Python/DDS interfaces.

Graph Neural Network & NLP & Contrastive Learning during M.S.

- To model different entities, such as text into graphs (both homogeneous and heterogeneous) and to use various types of graph neural networks to predict their properties and interactions.
- Transformer, text entity extraction, text searching alignment, relation construction, representation of tokens, similarity-based recommendation.
- Advanced contrastive learning, contrastive loss innovation during master.

Software/Service/System Implementation during FNii

- Robot service design and development.
- Web development, backend and front end, one project still in service (for 5 years).
- Implementation of federated learning platform via FATE core service, architecture, etc.

Industry

Research Engineer, Full Time, CUHK(Shenzhen)-FNii, Shenzhen July 2023 – present

- Vision Language Navigation and (Robot Arm) Vision Language Action.

NLP Algorithm Intern, Kuaishou, Beijing May 2021 – June 2022

- Text searching alignment by designed transformers and contrastive learning.

Algorithm Engineer Intern, IDEA, Shenzhen June 2022 – Oct 2022

- 2 transferrable graph embedding (Graph Sage and random walk) experiments, see GitHub repo1 and repo2. Implementation of PPRGO.

Open Source & Projects

Replicate and Deploy VLA/VLN Projects 2024

- VLN replication such as GoToAnything in both Habitat and OmniGibson, which includes intrinsic parameter, point cloud, coordinates calibration, HTTP communication build-up.
- VLA transfer such as OpenVLA (CoRL'24), ReKep (ICRA'25) in OmniGibson, which includes end-effector controlling, locomotion controlling, object locating, project refactoring.
- Open-sourced easy-to-use keyboard human control capture system and nearest frontier exploration example in Habitat3.

PyTorch Geometric (PyG) Contributor & Chinese Poetry Dataset check PyG check dataset

- Implementation of DMGI to PyG (a popular GNN learning library).
- The dataset got hundreds stars from volunteers.
- I regularly post technical reports and articles on CSDN, which have accumulated over 1 million views.

Program Lanugage/Operating System Skills

Languages: Python, ROS, bash, C++, Latex, Git, SQL, Linux, Windows